

2 Collagen is a fibrous protein that is found in many tissues in animals.

(a) Describe the structure of a collagen molecule **and** the structure of a collagen fibre.

collagen molecule

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collagen fibre

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[5]

(b) Some amino acids in collagen can be modified to improve the stability of the protein. For example, the amino acid lysine can be modified to form hydroxylysine.

Fig. 2.1 shows a disaccharide bonded to the amino acid hydroxylysine in a collagen molecule. The disaccharide is made from two monosaccharides, which are indicated by the labels **D** and **E** in Fig. 2.1.

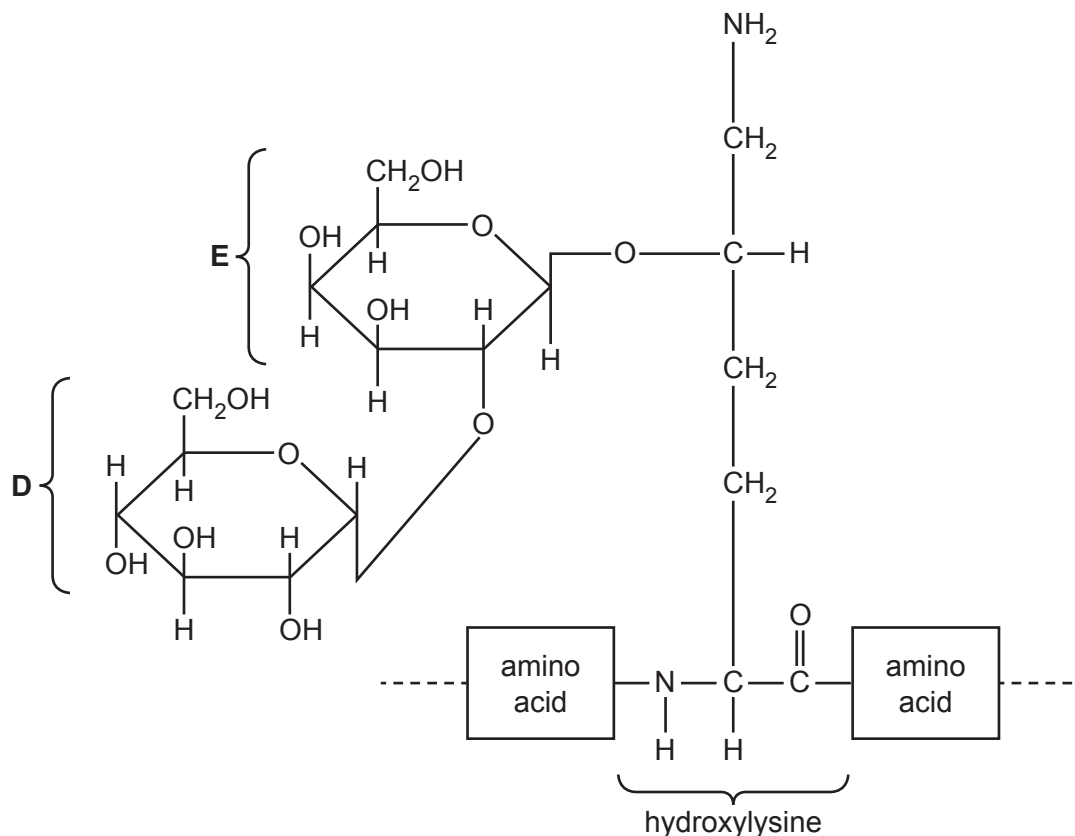


Fig. 2.1

(i) Give the precise name of the monosaccharide labelled **D** in Fig. 2.1.

..... [1]

(ii) On Fig. 2.1, label the glycosidic bond with the letter **G**.

Write your answer on Fig. 2.1. [1]

(iii) On Fig. 2.1, draw a circle around the R group of hydroxylysine and label the circle with the letter **R**.

Write your answer on Fig. 2.1. [1]

(c) Osteogenesis imperfecta is a disease that results from a deficiency in collagen.

Suggest how a **named** tissue or structure is affected in a person who has this disease.

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..... [2]

[Total: 10]